Coalitional monotonicity and the core: minimal impossibility, an exact-games escape, and the boundary of surplus-distributor solutions

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July 2026

Abstract

We study coalitional monotonicity of single-valued core selections in transferable-utility (TU) games. First, we pin down the minimal size of an impossibility certificate for the classical incompatibility of core selection and coalitional monotonicity: three-game certificates (a base game and two single-coalition increases) appear already in the textbook proof of Peleg and Sudhölter, and we record the complementary elementary fact that two-game families are always feasible—so three is exactly minimal—together with a self-contained three-game certificate admitting a six-line proof. Second, we map the monotonicity boundary of the Arin–Katsev surplus-distributor (SD) prenucleolus: coalitional monotonicity fails already on *exact* games ($n = 4$, a minimized witness), the SD-prekernel is multi-valued on exact games, population monotonicity fails even on convex games ($n = 3$, minimal), and on permutation games the SD solution amplifies the nucleolus's failure of pairwise monotonicity, whereas on assignment games extensive searches support monotonicity of both solutions. Third, we study whether the impossibility descends to exact games. For $n \leq 4$ it does not: by known results on large cores and egalitarian solutions, a coalitionally monotonic core selection exists on four-player exact games. The Housman–Clark mechanism is provably unavailable on exact games of any size, common-core intersection of bumped exact games can nevertheless fail (a four-arm example whose family remains feasible), and by a compactness argument existence for each n is *equivalent* to feasibility of all finite bump families, which we conjecture for $n \geq 5$.

Keywords: cooperative games; core; coalitional monotonicity; nucleolus; SD-prenucleolus; exact games.

1 Introduction

A transferable-utility (TU) cooperative game assigns to each coalition the worth it can guarantee on its own; a single-valued solution recommends one division of $v(N)$. Two desiderata have organized much of the literature on such solutions. *Core selection*: the recommendation should be stable against coalitional deviation whenever stability is possible. *Coalitional monotonicity*

*A machine-readable certificate supplement with a standalone one-command verifier accompanies this paper as Online Resource 1; see Section 8 for the verification methodology and the disclosure of AI assistance.

(Young 1985): if exactly one coalition's worth rises, no member of that coalition should receive less.

The two desiderata are incompatible. Young (1985) proved that for $|N| \geq 5$ no core selection is coalitionally monotonic; Housman and Clark (1998) extended the impossibility to $|N| \geq 4$ and showed it is tight ($n = 3$ admits an infinite family of monotonic core selections). The nucleolus fails coalitional monotonicity even on four-player convex games (Hokari 2000), as do the per-capita nucleolus (Arin 2013) and the core-center (Mirás Calvo et al. 2021). Against this background, Arin and Katsev (2011, 2014, 2016a, 2018) introduced the *surplus-distributor prenucleolus* (SD-prenucleolus),[1] a lexicographic core selection that *is* coalitionally monotonic on convex games and on monotonic games with veto players. On the domain of convex games alone, egalitarian solutions in the tradition of Dutta and Ray (1989) are also coalitionally monotonic core selections (Dietzenbacher 2021); what distinguishes the SD-prenucleolus is that it selects from the core on *all* balanced games while retaining monotonicity on both the convex and the monotonic-veto domains. How far that good behavior extends beyond these two classes was, to our knowledge, open.

This paper answers that question and, in doing so, isolates what we believe is the right open problem in this area. Our contributions:

1. **Minimal impossibility (Section 3).** Three-game impossibility families (a base and two bumps) are already present in the literature: the textbook proof of Peleg and Sudhölter (2003, Theorem 8.2.3) has exactly this shape, and the contradiction in the construction of Arin (2013) likewise uses only a base and two of its bumps. What appears not to have been recorded is optimality: we prove that two-game families are always feasible, so three is *exactly* the minimal size of an impossibility certificate, and we give a self-contained three-game certificate whose infeasibility admits a six-line proof.

2. **The SD boundary (Sections 4–5).** The coalitional monotonicity of the SD-prenucleolus does not extend one step beyond convexity: it fails on *exact* games at $n = 4$ (convex $\subsetneq$ exact $\subsetneq$ totally balanced $\subsetneq$ balanced), with a minimized integer witness. Three companion results complete the map: the SD-prekernel is multi-valued on exact games (containing a positive-dimensional core segment, so the convex-domain uniqueness theorem of Arin and Katsev (2018) is tight); the SD-prenucleolus is *not* population monotonic even on convex games (witness at the minimal size $n = 3$, with a short hand proof); and on permutation games the SD-prenucleolus fails pairwise monotonicity on the same witness as the classical nucleolus, with a larger violation. Certificates for the computational theorems appear in Appendix A.

3. **The exact-games question (Section 7).** The Young/Housman–Clark impossibility does not descend to exact games with at most four players: combining the large-core result of Biswas et al. (1999) with Dietzenbacher's monotonic egalitarian core selection on PES-stable games (Dietzenbacher 2021) yields a coalitionally monotonic core selection on $\mathcal{E}_4$ (Proposition 7.8). For general n the question is open, and we show the known impossibility mechanisms cannot decide it: the singleton-core pinning mechanism cannot exist on exact games (Lemma 7.1); face bounds of groups are tight and invariant along monotonicity chains of proper bumps (Theorem 7.2); overlapping two-arm families always admit a

[1] The literature uses both "SD-nucleolus" (e.g. the survey of Iñarra et al. 2020) and "SD-prenucleolus" (the titles of Arin and Katsev 2011, 2014) for the Arin–Katsev construction; we follow the latter. See Section 4.

common core point (Lemma 7.3); yet common-core intersection can fail at $m = n = 4$ arms (Proposition 7.5) while the associated family remains feasible—so family feasibility, not core intersection, is the right invariant. A compactness argument (Proposition 7.6) shows that for each n, existence of a coalitionally monotonic core selection on exact games is *equivalent* to the feasibility of all finite bump families, which we conjecture for $n \geq 5$ (Conjecture 7.9). The procedural egalitarian route cannot establish this conjecture on all exact games: Mauri Masdeu (2017, Example 5) already gives an exact five-player game that is not PES-stable.

4. **Market games (Section 6).** The nucleolus is *not* pairwise monotonic on permutation games: raising a single entry of the underlying value matrix can strictly lower the payoff of an agent named by that entry (smallest witness found: $n = 4$, integer matrix of total mass 20). In contrast, on assignment games—the two-sided special case—we found no violation for the nucleolus or the SD-prenucleolus in an extensive float-screened search that includes one fully exact component, an exhaustive certified sweep of a 2×2 grid. For the nucleolus this independently supports a theorem announced in the 2008 OTKA project report of Solymosi, Forgó and Pintér (Solymosi et al. 2008), whose proof we have not been able to access; for the SD-prenucleolus the statement appears new and we pose it as a conjecture. The permutation witness shows any proof must use two-sidedness itself.

All positive computational claims in this paper (witnesses, solution values, class memberships) are certified in exact rational arithmetic; lexicographic solutions are certified by Kohlberg-type balancedness criteria and, for headline witnesses, re-derived by an algorithmically independent exact solver. Negative search outcomes are float-screened, as detailed in Section 8; Appendix A lists human-checkable certificates, reproducible with the supplement's standalone verifier.

2 Preliminaries

A *TU game* is a pair (N, v) with $N = \{1, \ldots, n\}$ and $v : 2^N \to \mathbb{R}$, $v(\emptyset) = 0$. For $x \in \mathbb{R}^N$ and $S \subseteq N$ write $x(S) = \sum_{i \in S} x_i$. The *core* is $\mathrm{Core}(v) = \{x : \ x(N) = v(N), \ x(S) \geq v(S) \ \forall S\}$; v is *balanced* if $\mathrm{Core}(v) \neq \emptyset$, *totally balanced* if every subgame is balanced, *exact* (Schmeidler 1972) if for every S there is $x \in \mathrm{Core}(v)$ with $x(S) = v(S)$, and *convex* if $v(S \cup T) + v(S \cap T) \geq v(S) + v(T)$ for all S, T. It is standard that convex $\Rightarrow$ exact $\Rightarrow$ totally balanced $\Rightarrow$ balanced, all inclusions strict for $n \geq 4$ (respectively $n \geq 3$ for the last).

A collection $\mathcal{B}$ of nonempty proper coalitions is *balanced* if there are strictly positive weights $(\lambda_S)_{S \in \mathcal{B}}$ with $\sum_{S \in \mathcal{B}} \lambda_S \mathbf{1}_S = \mathbf{1}_N$. A *core selection* on a class $\mathcal{C}$ of balanced games is a map θ with $\theta(v) \in \mathrm{Core}(v)$ for all $v \in \mathcal{C}$.

Definition 2.1 (Coalitional monotonicity (Young 1985; Housman and Clark 1998))**.** *A single-valued solution θ is* coalitionally monotonic *on $\mathcal{C}$ if for all $u, v \in \mathcal{C}$ with $u(T) \leq v(T)$ for some coalition T and $u(S) = v(S)$ for all $S \neq T$, we have $\theta_i(u) \leq \theta_i(v)$ for all $i \in T$.*

Aggregate monotonicity is the special case $T = N$. *Population monotonicity* on a class closed under subgames requires $\theta_i(v|_S) \leq \theta_i(v|_T)$ for all $i \in S \subseteq T$; it suffices to check $|T \setminus S| = 1$.

Throughout, a *bump* of v at (S, d), $d > 0$, with $\emptyset \neq S \subseteq N$, is the game $v + d \cdot e_S$ agreeing with v everywhere except $(v + d \cdot e_S)(S) = v(S) + d$; the bump is *proper* if $S \neq N$. A *(bump) family* is a finite set of games together with bump relations between them; a family is *feasible* if one can choose a core element for each game satisfying all the monotonicity inequalities of

Definition 2.1 along the bump relations. The existence of a monotonic core selection on a class immediately implies every family inside the class is feasible; Proposition 7.6 states the converse.

We will use repeatedly the following elementary consequence of exactness.

Lemma 2.2. *Let v be exact and $T \subseteq N$ nonempty. Then $\max_{x\in\mathrm{Core}(v)} x(T) = v(N) - v(N\setminus T)$. Moreover, for any partition $\{T_1,\dots,T_m\}$ of N with $m \geq 2$, $\sum_k \big(v(N) - v(N\setminus T_k)\big) \geq v(N)$; for $m = 1$ the same inequality holds trivially with equality.*

Proof. For $x \in \mathrm{Core}(v)$, $x(T) = v(N) - x(N\setminus T)$, so maximizing $x(T)$ is minimizing $x(N\setminus T)$; by exactness the minimum equals $v(N\setminus T)$. For the second claim with $m \geq 2$, the collection $\{N\setminus T_k\}_k$ is balanced with weights $1/(m-1)$, so balancedness of v gives $\sum_k v(N\setminus T_k) \leq (m-1)\, v(N)$, which rearranges to the claim. □

Lemma 2.3 (exact bump arcs)**.** *Let v be exact and let S be nonempty. If $S \subsetneq N$ and $v' = v + de_S$ is exact for some $d > 0$, then $v + te_S$ is exact for every $0 \leq t \leq d$. If $S = N$, then $v + te_N$ is exact for every $t \geq 0$.*

Proof. First let $S \subsetneq N$ and write $C_t = \mathrm{Core}(v) \cap \{x : x(S) \geq v(S) + t\}$. This is the core of $v + te_S$ and contains $\mathrm{Core}(v')$. Exactness of v supplies a point of its core with $x(S) = v(S)$, while any point of $\mathrm{Core}(v')$ has $x(S) \geq v(S) + d$. Convexity therefore gives a point of C_t tight at S. For $T \neq S$, the function

$$f_T(t) = \min_{x\in C_t} x(T)$$

is nondecreasing, is at least $v(T)$, and equals $v(T)$ at both endpoints by exactness of v and v'. Hence it is identically $v(T)$.

Now let $S = N$. For each proper T, choose $x^T \in \mathrm{Core}(v)$ with $x^T(T) = v(T)$ and a player $j \notin T$. Then $x^T + te_j$ belongs to the core of $v + te_N$ and remains tight at T. Efficiency handles N itself. □

Thus every bump edge inside the exact class represents a whole exact one-parameter arc, not merely a relation between two isolated games. This permits local or parametric selection arguments to be integrated along the same edges that define the finite-family feasibility problem.

3 Minimal impossibility on balanced games

Housman and Clark (1998) prove that no core selection on four-player balanced games is coalitionally monotonic, using a family of five games (a base and four bumps). Certificates with three games are in fact already present in the literature: the textbook proof of Peleg and Sudhölter (2003, Theorem 8.2.3) uses precisely a four-player base game with two single-coalition increases, and the contradiction in the construction of Arin (2013) also needs only a base and two of its bumps. The observation that appears to be new here is the matching lower bound: three games cannot be improved upon.

Lemma 3.1. *Every two-game family $\{v,\, v + d\cdot e_S\}$ of balanced games is feasible.*

Proof. If $S \neq N$, then $\mathrm{Core}(v + d\cdot e_S) \subseteq \mathrm{Core}(v)$, since the bump only tightens the constraint at S; pick any $x \in \mathrm{Core}(v + d\cdot e_S)$ and assign it to both games. If $S = N$, pick $x \in \mathrm{Core}(v)$ and assign $x + \frac{d}{n}\mathbf{1}$ to the bumped game: it is efficient for it, satisfies every core constraint (each $x(T)$ only increases), and dominates x coordinatewise. In both cases all monotonicity inequalities hold. □

Theorem 3.2. *Let $N = \{1,2,3,4\}$ and define the balanced game v by*

$$v(\{1,3\}) = v(\{2,3\}) = v(\{1,4\}) = v(\{2,4\}) = v(\{3,4\}) = 3,$$
$$v(\{1,2,3\}) = v(\{1,2,4\}) = v(\{1,3,4\}) = 1, \quad v(\{2,3,4\}) = 5, \quad v(N) = 6,$$

all other coalitions 0. *Let $v_A = v + 3 \cdot e_{\{1,2,3\}}$ and $v_B = v + 1 \cdot e_{\{2,3,4\}}$; all three games are balanced, as witnessed by $(1,1,2,2) \in \mathrm{Core}(v) \cap \mathrm{Core}(v_A)$ and $(0,0,3,3) \in \mathrm{Core}(v_B)$. Then the family $\{v, v_A, v_B\}$ is infeasible: no choice of core elements $x \in \mathrm{Core}(v)$, $y \in \mathrm{Core}(v_A)$, $z \in \mathrm{Core}(v_B)$ satisfies $x_i \le y_i$ $(i \in \{1,2,3\})$ and $x_i \le z_i$ $(i \in \{2,3,4\})$. Consequently no coalitionally monotonic core selection exists on any class of games containing these three games; by Lemma 3.1, three is the minimal possible size of such a certificate.*

Proof. In $\mathrm{Core}(v_A)$: adding $y(\{1,2,3\}) \ge 4$, $y(\{1,4\}) \ge 3$ and $y(\{2,4\}) \ge 3$ gives $2y_1 + 2y_2 + y_3 + 2y_4 \ge 10$, i.e. $2\,v_A(N) - y_3 \ge 10$, so $y_3 \le 2$. In $\mathrm{Core}(v_B)$: adding $z(\{1,3\}) \ge 3$, $z(\{1,4\}) \ge 3$ and $z(\{2,3,4\}) \ge 6$ gives $2z_1 + z_2 + 2z_3 + 2z_4 \ge 12$, so $z_2 \le 0$. Monotonicity then forces $x_3 \le y_3 \le 2$ and $x_2 \le z_2 \le 0$, whence $x(\{2,3\}) \le 2 < 3 = v(\{2,3\})$, contradicting $x \in \mathrm{Core}(v)$. Minimality follows from Lemma 3.1. □

Remark 3.3. *The proof is a rearrangement of an exact Farkas certificate with eleven multipliers (nine core and monotonicity constraints with weight* 1*; the two efficiency identities with weight* 2*), found by linear programming over the family (Section 8; the multiplier table is in the supplement). None of the three games is exact, consistent with the obstructions of Section 7.*

4 The SD-prenucleolus

4.1 Definition

Fix (N, v) and an efficient x $(x(N) = v(N))$. The SD-prenucleolus of Arin–Katsev (Arin and Katsev 2011, 2014) is defined through a recursive *satisfaction* vector, as follows. Let $\mathcal{C} = 2^N \setminus \{\emptyset, N\}$. Maintain a set $H \subseteq \mathcal{C}$ of coalitions whose satisfactions $F(T;x)$ have been fixed. For $S \notin H$ let $H(S) = \bigcup\{T \in H : \ T \subseteq S\}$ (a set of *players*) and, for $i \in H(S)$,

$$f(i,S) \ = \ \min\{F(T;x) : \ T \in H, \ i \in T \subseteq S\}.$$

The current candidate satisfaction of S is

$$F_H(S) = \begin{cases} \dfrac{x(S) - v(S) - \sum_{i \in H(S)} f(i,S)}{|S| - |H(S)|}, & H(S) \neq S, \\ \\ x(S) - v(S) - \sum_{i \in S} f(i,S) + \max_{i \in S} f(i,S), & H(S) = S. \end{cases}$$

Starting from $H = \emptyset$ (all candidates are then per-capita satisfactions $(x(S) - v(S))/|S|$), repeatedly fix *all* coalitions attaining the minimal current candidate at that value and update; the procedure terminates with the satisfaction vector $F(\cdot\,; x)$. The *SD-prenucleolus* $\mathrm{SD}(N, v)$ is the efficient x whose non-decreasingly sorted satisfaction vector is lexicographically maximal. It is single-valued, continuous, covariant, a core selection on balanced games, strongly aggregate monotonic, and coalitionally monotonic on convex games and on monotonic veto games (Arin and Katsev 2011, 2014, 2016a, 2018).

The solution admits a characterization in the style of Kohlberg (1971) (Arin and Katsev 2011, 2018): an efficient x equals $\mathrm{SD}(N, v)$ if and only if for every threshold $\alpha \in \mathbb{R}$ the lower level set $\{S \in \mathcal{C} : \ F(S;x) \le \alpha\}$ is empty or balanced. We use this as the certification instrument throughout.

4.2 Implementation and validation

We compute $\mathrm{SD}(N, v)$ by a Maschler-type successive linear-programming scheme (at each stage the candidate satisfactions of unfixed coalitions are affine in x, because fixed satisfactions are constants), with every output certified by (i) an exact evaluation of the recursion above at the candidate point and (ii) the Kohlberg-type criterion, with balancedness of each level set established by an exact linear program producing strictly positive weights. As additional regression checks (not proofs), headline witnesses were probed for lexicographic optimality along hundreds of exact rational perturbation directions.

The implementation reproduces exactly: the worked example of Arin and Katsev (2018); the prenucleolus on two-player games; the serial rule on random monotonic veto games (Theorem 14.3 of Arin and Katsev 2018, using the explicit serial-rule formula of Arin and Feltkamp 2012); the minimal-overlap division on the bankruptcy game $(E; c) = (200; 100, 200, 300)$, namely (100/3, 250/3, 250/3) (Theorem 14.4 of Arin and Katsev 2018), which differs from the Talmud/nucleolus division $(50, 75, 75)$; strict strong aggregate monotonicity on 150 random games; and coalitional monotonicity along 8 random convex bumps. On the five-player benchmark game of Arin–Iñarra reported in Iñarra et al. (2020) our classical nucleolus solver reproduces the published value $(16/9, 16/9, 16/9, 4/3, 4/3)$ exactly.

5 The boundary of the SD-prenucleolus

5.1 Coalitional monotonicity fails on exact games

Theorem 5.1. *The SD-prenucleolus is not coalitionally monotonic on the class of exact games, already for $n = 4$. Witness: let v be*

$$v(\{2,3\}) = 1,\ v(\{1,2,3\}) = 1,\ v(\{2,4\}) = 13,\ v(\{1,2,4\}) = 15,$$

$$v(\{1,3,4\}) = 14,\ v(\{2,3,4\}) = 13,\ v(N) = 30,$$

all other coalitions 0*, and let $v' = v + 1 \cdot e_{\{1,3,4\}}$. Both games are exact (hence totally balanced; neither is convex). Then*

$$\mathrm{SD}(v) = \left(\tfrac{53}{12},\ \tfrac{19}{4},\ \tfrac{15}{4},\ \tfrac{205}{12}\right), \qquad \mathrm{SD}(v') = \left(\tfrac{35}{8},\ \tfrac{33}{8},\ \tfrac{33}{8},\ \tfrac{139}{8}\right),$$

so player $1 \in \{1, 3, 4\}$ loses $1/24$ when the worth of a coalition containing her rises.

Proof (computer-assisted). Exactness of each game is certified by $2^n - 2$ exact linear programs exhibiting, for every coalition S, a core element with $x(S) = v(S)$. That the displayed points are the SD-prenucleoli follows from the Kohlberg-type criterion of Section 4: the full satisfaction level structure of both points and strictly positive balancing weights for the lower level sets are listed in Appendix A and can be checked by hand. The payoff comparison is exact rational arithmetic. □

The witness was found by search and greedily minimized (total mass 87, seven nonzero values); both games are moreover superadditive, so the failure holds on exact (hence totally balanced) superadditive games. Since Arin–Katsev prove monotonicity on convex games, the boundary along convex $\subsetneq$ exact $\subsetneq$ totally balanced $\subsetneq$ balanced is *convexity itself.*

Corollary 5.2. *Exactness does not imply SD-relevance, the domain hypothesis of the general monotonicity theorem of Arin and Katsev (2018).*[2]

5.2 The SD-prekernel is multi-valued on exact games

Arin and Katsev (2016b, 2018) define the SD-prekernel through bilateral complaints $s_{ij}(x) = \min\{F(S;x) : i \in S,\ j \notin S\}$: it is the set of efficient x with $s_{ij}(x) = s_{ji}(x)$ for all $i \neq j$. It contains the SD-prenucleolus, and on convex games it equals it (a singleton).

Theorem 5.3. *The SD-prekernel can be multi-valued—indeed positive-dimensional—on exact games. For the exact, totally balanced, non-convex game*

$$v(\{1\}) = 5,\ v(\{2\}) = 6,\ v(\{3\}) = 2,\ v(\{4\}) = 5,$$
$$v(\{1,2\}) = 20,\ v(\{1,3\}) = 10,\ v(\{2,3\}) = 8,\ v(\{1,4\}) = 12,\ v(\{2,4\}) = 14,\ v(\{3,4\}) = 12,$$
$$v(\{1,2,3\}) = 22,\ v(\{1,2,4\}) = 25,\ v(\{1,3,4\}) = 17,\ v(\{2,3,4\}) = 18,\quad v(N) = 32,$$

the SD-prekernel contains the segment $\mathrm{SD}(v) + t\,(1,-1,-1,1)$, $t \in [-1/2, 1/2]$, *where* $\mathrm{SD}(v) = (19/2,\ 21/2,\ 9/2,\ 15/2)$*; every point of the segment lies in the core and equalizes all six bilateral complaints exactly.*

Proof. Write $x^t = \mathrm{SD}(v) + t\,(1,-1,-1,1)$ for $t \in [-1/2, 1/2]$. Core membership of x^t reduces to that of the endpoints $(9, 11, 5, 7)$ and $(10, 10, 4, 8)$, which is a direct check against the fifteen constraints. Evaluating the recursion of Section 4 at x^t gives, for every $t \in [-1/2, 1/2]$,

$$F_{\{1,2\}} = F_{\{3,4\}} = 0, \qquad F_{\{1,3\}} = F_{\{2,4\}} = 2,$$
$$F_{\{3\}} = F_{\{1,2,3\}} = \tfrac{5}{2} - t, \qquad F_{\{4\}} = F_{\{1,4\}} = F_{\{1,2,4\}} = \tfrac{5}{2} + t,$$
$$F_{\{2\}} = F_{\{2,3\}} = F_{\{2,3,4\}} = \tfrac{9}{2} - t, \qquad F_{\{1\}} = F_{\{1,3,4\}} = \tfrac{9}{2} + t$$

The displayed functions are proved by running the recursion symbolically, stage by stage, on each open half-interval; the fixing order is constant on each. For $t \in (-\frac{1}{2}, 0)$ the stages are

$$\{12,34\};\ \{13,24\};\ \{4,14,124\};\ \{3,123\};\ \{1,134\};\ \{2,23,234\}$$

at levels 0, 2, $\frac{5}{2} + t$, $\frac{5}{2} - t$, $\frac{9}{2} + t$, $\frac{9}{2} - t$ respectively, and for $t \in (0, \frac{1}{2})$ they are

$$\{12,34\};\ \{13,24\};\ \{3,123\};\ \{4,14,124\};\ \{2,23,234\};\ \{1,134\}$$

at levels 0, 2, $\frac{5}{2} - t$, $\frac{5}{2} + t$, $\frac{9}{2} - t$, $\frac{9}{2} + t$. At each stage, direct substitution of x^t into the two-case formula of Section 4, using the already-fixed values, yields exactly the displayed candidates and confirms that the stated minimizers are fixed next (the required inequalities, e.g. $2 \leq \frac{5}{2} \mp t$ and $\frac{5}{2} + t < \frac{5}{2} - t$ for $t < 0$, hold strictly on the open half-intervals). The boundary values $t \in \{-\frac{1}{2}, 0, \frac{1}{2}\}$ are verified separately by direct exact computation (supplement), where some stages merge but the resulting satisfaction values still obey the displayed formulas. Reading off the bilateral complaints $s_{ij} = \min\{F_S : i \in S \not\ni j\}$: for $|t| \leq \frac{1}{2}$, $s_{12} = s_{21} = 2$ (attained at $\{1,3\}$ resp. $\{2,4\}$), $s_{34} = s_{43} = 2$, and all other ordered pairs have $s_{ij} = 0$ (attained at $\{1,2\}$ or $\{3,4\}$). All six equalities $s_{ij} = s_{ji}$ therefore hold along the segment. That the midpoint is $\mathrm{SD}(v)$ is certified by the Kohlberg-type criterion (Appendix A). □

[2] We refer to Arin and Katsev (2018) for the definition of SD-relevant games; here we use only the fact that their theorem establishes coalitional monotonicity of the SD-prenucleolus on that class, which by Theorem 5.1 therefore cannot contain both witness games.

Thus the uniqueness theorem for convex games is tight, with the same boundary as Theorem 5.1. (Perturbed control points off the segment fail the equalities, confirming the checker discriminates.)

5.3 Population monotonicity fails on convex games

Theorem 5.4. *The SD-prenucleolus is not population monotonic on convex games, already at $n = 3$, which is minimal: for $n \le 2$ every core selection is population monotonic, since individual rationality gives $\theta_i(v|_{\{i\}}) = v(\{i\}) \le \theta_i(v|_T)$ for any $T \ni i$. For the convex, exact game $v(\{1,2\}) = v(\{2,3\}) = 1$, $v(N) = 2$ (all else 0):*

$$\mathrm{SD}\big(v|_{\{2,3\}}\big) = \big(\tfrac{1}{2}, \tfrac{1}{2}\big), \qquad \mathrm{SD}(v) = \big(\tfrac{1}{3},\ \tfrac{4}{3},\ \tfrac{1}{3}\big),$$

so player 3 falls from 1/2 to 1/3 when player 1 joins.

Proof. Convexity is immediate. The two-player subgame on $\{2,3\}$ has $v(\{2\}) = v(\{3\}) = 0$, $v(\{2,3\}) = 1$; the SD-prenucleolus coincides with the prenucleolus on two-player games, giving the equal split $(1/2, 1/2)$.

For the full game we verify $x = (1/3, 4/3, 1/3)$ by the Kohlberg-type criterion. The recursion finishes in two stages (the five coalitions at the minimal level are fixed first, then $\{2\}$): the per-capita satisfactions are $F(\{1\}) = F(\{3\}) = 1/3$; $F(\{1,2\}) = F(\{2,3\}) = (x(S) - 1)/2 = 1/3$; $F(\{1,3\}) = x(\{1,3\})/2 = 1/3$; $F(\{2\}) = 4/3$. The lower level sets are: for $\alpha \in [1/3, 4/3)$, $\mathcal{B}_1 = \{\{1\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}\}$, balanced with strictly positive weights $\lambda_{\{1,2\}} = \lambda_{\{2,3\}} = \frac{1}{2}$, $\lambda_{\{1,3\}} = \frac{1}{3}$, $\lambda_{\{1\}} = \lambda_{\{3\}} = \frac{1}{6}$ (player 1: $\frac{1}{2}+\frac{1}{3}+\frac{1}{6} = 1$; player 2: $\frac{1}{2}+\frac{1}{2} = 1$; player 3: $\frac{1}{3}+\frac{1}{2}+\frac{1}{6} = 1$); for $\alpha \ge 4/3$, the set of all proper coalitions, which is balanced. Hence every lower level set is empty or balanced and $x = \mathrm{SD}(v)$. Finally $1/2 > 1/3$. □

In exploratory float-screened sampling of convex games (integer-valued supermodular games generated as nonnegative integer combinations of unanimity games), violations appeared in roughly half of the instances at $n = 4$ and more at $n = 5$; we report these frequencies as indicative only, since the exploratory counting runs were not preserved as seeded scripts. The phenomenon is in any case common under such sampling, not exceptional. The SD-prenucleolus therefore separates the two monotonicities on its home class: it is coalitionally monotonic there (Arin and Katsev 2016a) but not population monotonic—in contrast with the egalitarian solution of Dutta and Ray (1989), which is population monotonic on convex games.

6 Market games: pairwise monotonicity

An *assignment game* (Núñez and Rafels 2003) is generated by a nonnegative matrix $A = (a_{ij})$ over two disjoint sides M (rows) and M' (columns): the player set is $M \cup M'$ and $v(S)$ is the maximum weight of a matching between $S \cap M$ and $S \cap M'$ with weights a_{ij}. A *permutation game* (Tijs et al. 1984) is generated by a square matrix $K = (k_{ij})$ over a single player set: $v(S) = \max_\pi \sum_{i \in S} k_{i\pi(i)}$, the maximum over permutations π of S. Assignment games embed into permutation games (Curiel and Tijs 1986); both classes are totally balanced.

Pairwise monotonicity (Núñez and Rafels 2003) is the natural monotonicity axiom for such games: if one entry of the matrix rises, neither of the two agents named by that entry may lose. The buyers-optimal, sellers-optimal and τ-value rules satisfy it on assignment games (Núñez and Rafels 2003; van den Brink et al. 2021); Miquel (2009) constructs a bespoke pairwise-monotonic

core selection for permutation games.[3] For the nucleolus on assignment games, the 2008 OTKA project report of Solymosi, Forgó and Pintér (Solymosi et al. 2008) announces: "We proved that the nucleolus is pairwise-monotonic on assignment games"; the project's 2012 final report attributes the result to an unpublished manuscript, "Pairwise monotonicity of the nucleolus in assignment games" by T. Solymosi, M. Brugueras and T. E. S. Raghavan (2012), with a simplified proof in a revised version.[4] We have not been able to access either manuscript; our computations below independently support the statement.

Theorem 6.1. *The nucleolus is not pairwise monotonic on permutation games. Let K be the value matrix (rows: owners; columns: recipients)*

$$K = \begin{pmatrix} 0 & 7 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 9 & 1 & 0 \end{pmatrix},$$

and raise K_{43} from 1 to 2 (which raises only $v(\{3,4\})$ and $v(\{1,3,4\})$, each by one, and fixes $v(N) = 11$). The nucleolus moves from $(\frac{1}{2}, \frac{33}{4}, \frac{1}{2}, \frac{7}{4})$ to $(\frac{2}{3}, \frac{23}{3}, \frac{1}{3}, \frac{7}{3})$: player 3, a member of both improved coalitions, falls from $1/2$ to $1/3$. Both games are totally balanced and both nucleoli lie in the core. Moreover the SD-prenucleolus fails on the same instance, falling from $1/2$ to $1/4$ for the same player.

Proof (computer-assisted). Both nucleolus claims are certified by Kohlberg's criterion in exact arithmetic; Appendix A lists the excess level structure and balancing weights at the top level for both games, and the machine-readable supplement contains the full level data. The SD claims are certified as in Section 4.2. □

The witness is greedily minimized (integer matrix, total mass 20); a float-screened $n = 3$ sweep of 36,000 bumps found no candidate violation, so $n = 4$ is the smallest size at which we found the phenomenon. In this instance the improving pair's *joint* payoff rises ($9/4 \to 8/3$ for the nucleolus), so the witness does not violate Sasaki's weak (joint) pairwise monotonicity (Sasaki 1995); the bump's surplus is redistributed *within* the improving pair.

On assignment games, by contrast, an extensive search—float-screened random and tie-targeted 3×3, 4×3, 4×4 markets and hill-climbing on the violation margin, together with one fully exact component, an exhaustive Kohlberg-certified sweep of all 2×2 markets with entries in $\{0, \ldots, 4\}$ (2,500 bumps, zero violations, zero certification failures)—found no violation for either solution. For the nucleolus this is consistent with (Solymosi et al. 2008); for the SD-prenucleolus we know of no prior statement:

Conjecture 6.2. *The SD-prenucleolus is pairwise monotonic on assignment games.*

Conjecture 6.3. *The nucleolus of minimum-cost spanning tree games is cost monotonic in the sense of Dutta and Kar (2004) (raising one edge's cost never lowers an incident player's cost share).*

[3] We could not access the body of Miquel (2009); its abstract states a positive result for a merged-τ-value rule and makes no claim about the nucleolus. Whether an explicit nucleolus counterexample appears in its introduction is a caveat for the novelty (not the correctness) of Theorem 6.1.

[4] OTKA final report, `https://real.mtak.hu/12261/1/72856_ZJ1.pdf`.

Conjecture 6.3 is supported by 1,450 float-screened tests (with exact certification awaiting any candidate violation; none arose) at $n \in \{4, 5, 6\}$ including large bumps in the core-moving regime of Tan et al. (2025); the nucleolus is known to be monotonic on the fixed-tree subclass (Maschler et al. 2010), and Dutta–Kar's violating rule is Bird's, not the nucleolus. Theorem 6.1 shows any proof of the assignment-game statements must use two-sidedness: the property fails one class up.

7 Do exact games escape the impossibility?

The certificate games of Theorem 3.2, like those of Housman and Clark (1998), are not exact. This section first records that for $n \leq 4$ the existence question on exact games is settled affirmatively by known results (Proposition 7.8), then shows that the pinning and two-arm mechanisms underlying the known impossibility proofs are unavailable on exact games of any size, delimits the techniques by an explicit counterexample to a natural strengthening, and reduces the existence question to finite-family feasibility.

7.1 The pinning mechanism is unavailable

The Housman–Clark proof pins players via bumps whose cores collapse to singletons.

Lemma 7.1. *An exact game whose core is a singleton is additive. Consequently no exact game family can pin a player through a singleton core except trivially.*

Proof. If $\mathrm{Core}(v) = \{x\}$, exactness gives, for each S, a core element attaining $x(S) = v(S)$; but the core element is unique, so $x(S) = v(S)$ for every S, i.e. v is additive with $v(S) = \sum_{i \in S} x_i$. □

More generally, pinning arguments transmit *face bounds* $\max_{x \in \mathrm{Core}} x(T)$ along monotonicity chains. On exact games these bounds cannot contract under proper bumps:

Theorem 7.2 (Pin invariance under proper bumps)**.** *Let all games below be exact and let all bumps be proper. Fix a nonempty group $T \subseteq N$. Along any monotonicity chain that constrains T—i.e. any sequence of proper bumps at coalitions $S \supseteq T$—the face bound of Lemma 2.2 satisfies $\max_{x \in \mathrm{Core}} x(T) = v(N) - v(N \setminus T)$ in every game of the chain, with both $v(N)$ and $v(N \setminus T)$ unaltered by every bump in the chain. In particular, since face bounds over any partition sum to at least $v(N)$ (Lemma 2.2), no contradiction with efficiency can be derived from partition-aggregated face bounds, at any chain depth and for groups of any arity; we do not claim the analogue for arbitrary balanced collections of groups. (A bump at $S = N$ raises every face bound by d and is likewise harmless.)*

Proof. A proper bump at S constrains T only if $T \subseteq S$; since $T \cap (N \setminus T) = \emptyset$ and $S \neq N$, no such S equals $N \setminus T$ or N, so neither $v(N \setminus T)$ nor $v(N)$—and with them the face bound, by Lemma 2.2 applied in each (exact) game—changes along the chain. □

7.2 Two-arm families: positive results and their limits

Lemma 7.3 (Overlapping arms)**.** *Let b be exact and let $t_A = b + d_A e_A$, $t_B = b + d_B e_B$ be exact, where $A \neq B$ are nonempty proper coalitions with $A \cap B \neq \emptyset$. Then $M = b + d_A e_A + d_B e_B$ is balanced, and any $x^* \in \mathrm{Core}(t_A)$ maximizing $x(B)$ lies in $\mathrm{Core}(M) \subseteq \mathrm{Core}(b) \cap \mathrm{Core}(t_A) \cap \mathrm{Core}(t_B)$; assigning x^* to all three games shows the family $\{b, t_A, t_B\}$ is feasible.*

Proof. By exactness of t_A at $N \setminus B$ and $A \not\subseteq N \setminus B$ (as $A \cap B \neq \emptyset$),

$$\max_{x \in \mathrm{Core}(t_A)} x(B) = t_A(N) - t_A(N \setminus B) = b(N) - b(N \setminus B) = \max_{x \in \mathrm{Core}(b)} x(B) \;\geq\; b(B) + d_B,$$

the last inequality because $\mathrm{Core}(t_B) \neq \emptyset$ forces some core element of b with $x(B) \geq b(B) + d_B$. The maximizer x^* then satisfies every constraint of M. □

Proposition 7.4 (Disjoint arms: exactness chaining)**.** *Let b be exact and $t_A = b + d_A e_A$, $t_B = b + d_B e_B$ exact, where A, B are nonempty proper coalitions with $A \cap B = \emptyset$ and $R := N \setminus (A \cup B) \neq \emptyset$. Then*

$$b(N) \;\geq\; b(A) + d_A + b(B) + d_B + b(R),$$

i.e. the partition Farkas obstruction $\{A, B, R\}$ to the balancedness of $M = b + d_A e_A + d_B e_B$ vanishes. (This is one obstruction among many; the proposition does not by itself establish balancedness of M or feasibility of the family.)

Proof. Step 1. Every $x \in \mathrm{Core}(t_B)$ has $x(N \setminus A) = x(B) + x(R) \geq b(B) + d_B + b(R)$; since t_B is exact at $N \setminus A$ and $t_B(N \setminus A) = b(N \setminus A)$ (here $B \subsetneq N \setminus A$ because $R \neq \emptyset$, so the bump at B does not change the worth of $N \setminus A$), taking the minimum gives $b(N \setminus A) \geq b(B) + d_B + b(R)$. *Step 2.* By exactness of t_A at $N \setminus A$ there is $x^* \in \mathrm{Core}(t_A)$ with $x^*(N \setminus A) = b(N \setminus A)$, hence $x^*(A) = b(N) - b(N \setminus A)$; but $x^*(A) \geq b(A) + d_A$, so $b(N) \geq b(A) + d_A + b(N \setminus A)$. Chaining the two steps gives the claim. □

These techniques have a genuine limit, which the following example (due to a reviewer of an earlier draft) makes precise: common-core intersection of bumped games—the conclusion of Lemma 7.3—can fail when $m = n = 4$ arms are bumped, even though the associated monotonicity family remains feasible.

Proposition 7.5 (Core intersection can fail at $m = n = 4$; feasibility survives)**.** *Let $n = 4$, $b(N) = 5$ and $b(S) = 0$ for all proper S. For $k \in N$ let $t_k = b + 4\, e_{N \setminus \{k\}}$. Then b and each t_k are exact, and $\mathrm{Core}(t_k) = \{x \geq 0 : \; x(N) = 5, \; x_k \leq 1\}$, so $\bigcap_k \mathrm{Core}(t_k) = \emptyset$ (four inequalities $x_k \leq 1$ would force $x(N) \leq 4$); equivalently, the all-arm game $b + \sum_k 4\, e_{N \setminus \{k\}}$ is unbalanced. Nevertheless the star family $\{b, t_1, \ldots, t_4\}$ is feasible: take $\theta(b) = (2, 1, 1, 1)$, $\theta(t_1) = (1, \frac{4}{3}, \frac{4}{3}, \frac{4}{3})$, and for $k \neq 1$ the point of $\mathrm{Core}(t_k)$ obtained from $(2, 1, 1, 1)$ by moving one unit from player k to player 1 (e.g. $\theta(t_2) = (3, 0, 1, 1)$), which dominates $\theta(b)$ on $N \setminus \{k\}$.*

Proof. Exactness of b is clear (all mass can be placed on any single player). For t_k: the core is as displayed. For a proper $S \ni k$, the point with $x_k = 0$ and mass 5 on a player outside S attains $x(S) = 0 = t_k(S)$. For a proper $S \not\ni k$ with $S \neq N \setminus \{k\}$, there is a player $j \notin S \cup \{k\}$; the point with $x_k = 1$, $x_j = 4$ attains $x(S) = 0 = t_k(S)$. Finally the minimum of $x(N \setminus \{k\}) = 5 - x_k$ over the core is $4 = t_k(N \setminus \{k\})$. So t_k is exact. Emptiness of the intersection and unbalancedness (the collection $\{N \setminus \{k\}\}_k$ with weights $1/3$ gives $\frac{16}{3} > 5$) are immediate. Feasibility: each displayed $\theta(t_k)$ lies in $\mathrm{Core}(t_k)$ and dominates $\theta(b) = (2, 1, 1, 1)$ on every coordinate of $N \setminus \{k\}$. □

Proposition 7.5 corrects a conjecture we advertised in an earlier draft (that the all-arm game of exact single bumps is always balanced): the common-point method cannot decide the exact-games question, because family feasibility is strictly weaker than core intersection. Computationally, across more than 10^6 family-feasibility linear programs built from over 7,500 exact

four-player bases (two- and three-arm stars and depth-two lattices), floating-point screening flagged *no infeasible family of exact games*. The pipeline certifies any flagged infeasibility by an exact Farkas certificate (validated on the Housman–Clark family and on Theorem 3.2); the negative outcome itself is float-screen-level, and float screening can in principle produce false negatives near degeneracy. In light of Proposition 7.8 below, the four-player census outcome is in fact implied by the existence of a monotonic selection on $\mathcal{E}_4$, so we present it as validation of the feasibility pipeline rather than as evidence at the open frontier, which begins at $n = 5$.

7.3 Equivalence, and the state of the question

Proposition 7.6. *Fix n and let $\mathcal{E}_n$ be the class of exact n-player games. A coalitionally monotonic core selection exists on $\mathcal{E}_n$ if and only if every finite bump family within $\mathcal{E}_n$ is feasible.*

Proof. Necessity is immediate. For sufficiency, consider the product $P = \prod_{v \in \mathcal{E}_n} \mathrm{Core}(v)$, compact in the product topology by Tychonoff's theorem, since each core is a nonempty compact subset of $\mathbb{R}^n$. For each bump pair (u, v) in $\mathcal{E}_n$ let $C_{u,v} \subseteq P$ be the set of selections satisfying the (finitely many, closed) monotonicity inequalities of Definition 2.1 for that pair; $C_{u,v}$ is closed and depends on two coordinates only. A finite intersection of such sets involves finitely many games; feasibility of the corresponding finite family provides a point (extending arbitrarily elsewhere). By the finite intersection property the full intersection is nonempty, and any of its points is a coalitionally monotonic core selection. □

Proposition 7.7 (structure of a minimal finite obstruction)**.** *Suppose a finite bump family on q games is infeasible and is inclusion-minimal with respect to its games and bump relations. Label each directed bump edge by its bumped coalition. Then:*

(i) the underlying undirected graph is connected and the directed graph is acyclic;

(ii) for every edge $u \to w$ labeled S, some $i \in S$ has no alternative directed path from u to w all of whose edge labels contain i;

(iii) infeasibility is witnessed by at most $q(n-1)+1$ core or monotonicity halfspaces, after the q efficiency equalities are imposed. For rational games, the corresponding sparse Farkas certificate may be chosen rational.

Proof. If the undirected graph were disconnected, feasible selections for its components could be combined, contrary to minimal infeasibility. Along every directed edge exactly one coalition worth strictly increases, so the sum of all coalition worths is a strictly increasing potential; directed cycles are impossible.

For (ii), suppose instead that for every $i \in S$ there were such an alternative path. Transitivity of the monotonicity inequalities along that path would imply $\theta_i(u) \le \theta_i(w)$. Thus every inequality attached to the edge $u \to w$ would follow from the remaining edges, contradicting minimality.

Finally restrict the qn allocation variables to the product of the q efficiency hyperplanes, an affine space of dimension $q(n-1)$. Family feasibility is the intersection there of finitely many core and monotonicity halfspaces. Helly's theorem gives an empty subintersection of at most $q(n-1)+1$ halfspaces. With rational coefficients, the usual rational form of Farkas' lemma supplies rational multipliers on such a subfamily. □

Proposition 7.7 turns the search for a refuting exact family into a sparse labeled-DAG problem. In particular, disconnected stars, directed cycles, transitively redundant bump labels, and certificates whose support exceeds the Helly bound need not be searched when mining minimal obstructions.

For $n \leq 4$ the existence question is in fact already settled in the affirmative by combining two known results:

Recall that the core of v is *large* (Sharkey 1982) if for every $y \in \mathbb{R}^N$ with $y(S) \geq v(S)$ for all S there is $x \in \mathrm{Core}(v)$ with $x \leq y$, and call a game *PES-stable* if the procedural egalitarian solution of Dietzenbacher (2021) selects a core allocation of it.

Proposition 7.8 (Dietzenbacher 2021, with Biswas et al. 1999)**.** *A coalitionally monotonic core selection exists on the class $\mathcal{E}_4$ of exact four-player games (hence also on $\mathcal{E}_n$, $n \leq 4$).*

Proof. Every exact game with at most four players has a large core (Biswas et al. 1999). Large-core games are PES-stable, and on PES-stable games the procedural egalitarian solution is a single-valued core selection satisfying coalitional monotonicity (Dietzenbacher 2021); the combination is observed in Dietzenbacher (2021). (As a consistency check, we verified by exact vertex enumeration that all eight exact four-player games directly certified in this paper have large cores.) □

Proposition 7.6 frames both sides of the general ledger: the impossibility on balanced games *is* one infeasible finite family (Theorem 3.2); the results above are partial progress toward feasibility of all finite exact families for general n. The solutions studied in this paper fail coalitional monotonicity on exact games individually—the nucleolus (already on convex games (Hokari 2000)), the per-capita nucleolus (Arin 2013), the core-center (Mirás Calvo et al. 2021) (and, in our tests, a coordinate-extreme variant, in 136 of 1,059 exact bumps), and the SD-prenucleolus (Theorem 5.1)—while Proposition 7.8 shows that for $n \leq 4$ a different, egalitarian-type formula succeeds. For $n \geq 5$ no selection is known to work. Dietzenbacher's particular procedural egalitarian solution cannot supply one on all exact games: Mauri Masdeu (2017, Example 5) applies the same maximum-average procedure to an exact five-player game, obtains the zero vector, and observes that it lies outside the core. The recent follow-up of Dietzenbacher and Doğan (2025) develops egalitarianism under monotonicity constraints further but does not settle the existence of a different selection on exact games.

Conjecture 7.9. *For every fixed $n \geq 5$, a coalitionally monotonic core selection exists on the class $\mathcal{E}_n$ of exact n-player games.*

By Proposition 7.6, Conjecture 7.9 is equivalent to feasibility of all finite exact families for the given n. The pinning-based and partition-aggregated obstruction mechanisms are unavailable at any n (Lemma 7.1, Theorem 7.2); what is missing is a feasibility proof for arbitrary finite families—or a refuting family. Proposition 7.5 shows that one natural route to refutation, empty common intersection of bumped cores, is by itself insufficient. An argument for arbitrary finite families—or a new finite obstruction—is therefore still required.

8 Verification methodology

All results were produced by a search-then-certify discipline. Screening uses floating-point successive linear programs; every reported object is then reconstructed in exact rational arithmetic

and certified independently of the search path: nucleolus and SD points by Kohlberg-type balanced-collection criteria evaluated by an exact rational simplex method (written for this project after we found that a current computer-algebra system's exact linear-programming routine returns infeasible points as optima on three-variable instances; our solver is fuzz-tested against a standard floating-point solver with exact feasibility assertions), class memberships (balancedness, total balancedness, exactness, convexity) by exact linear programs, family infeasibility by exact Farkas certificates re-verified under constraint reordering, and headline witnesses additionally re-derived by an algorithmically independent exact successive-LP solver. Negative search outcomes (no violation found, no infeasible family found) are reported at the float-screen level: exact certification in those campaigns is triggered by candidate positives, and none arose; the sole fully exact negative sweep is the exhaustive 2×2 assignment grid. Known values from the literature (Talmud division, serial rule, minimal-overlap rule, a published five-player nucleolus, Hokari-type failures at their known minimal sizes, and the Housman–Clark family itself) are reproduced as anchors. The discipline caught nine false or overstated candidate results before any claim was circulated, including two floating-point artifacts near the exactness boundary and the overreaching core-intersection conjecture corrected in Proposition 7.5; we accordingly treat direct exact LP oracles as authoritative over derived closed forms. The computational search, the verification tooling, and the drafting of this paper made substantial use of large-language-model assistance; every mathematical claim was verified by the certification pipeline described here, and all text was reviewed by the author. Computations used Python 3.10 with exact arithmetic via the standard `fractions` module and floating-point screening via NumPy 2.2/SciPy 1.15 (HiGHS). A machine-readable supplement (`certificates.json`) contains, for every computational theorem, the full game tables, per-coalition exactness witnesses, complete satisfaction/excess tables, and strictly positive balancing weights for every cumulative level set, together with a one-command verifier (`verify_paper.py`) that re-checks all of it in exact arithmetic using only the Python standard library (it recomputes every satisfaction and excess table from scratch, checks that every stored level set equals the recomputed one, and validates the supplied balancing weights by pure rational arithmetic), and an environment manifest; together these form Online Resource 1.

9 Open problems

1. Prove (or refute) feasibility of every finite bump family of exact n-player games for $n \geq 5$; by Proposition 7.6 this settles Conjecture 7.9. Proposition 7.5 shows the common-core-point method cannot decide it, and the procedural egalitarian route is closed by the exact counterexample of Mauri Masdeu (2017, Example 5).

2. Recover the unpublished Solymosi–Brugueras–Raghavan proof (see Section 6) that the nucleolus is pairwise monotonic on assignment games, and prove Conjecture 6.2 for the SD-prenucleolus. Theorem 6.1 shows two-sidedness must be used.

3. Prove Conjecture 6.3 (Dutta–Kar cost monotonicity of the nucleolus on MCST games), known on the fixed-tree subclass (Maschler et al. 2010).

4. Characterize SD-relevance (Arin and Katsev 2018) within the exact games; Theorem 5.1 shows exactness is insufficient.

5. Determine whether the witness of Theorem 6.1 is pre-empted by an example in the introduction of Miquel (2009).

Statements and Declarations

Funding: none. **Data and code availability:** a machine-readable certificate supplement accompanies this paper as Online Resource 1 (`certificates.json`; `verify_paper.py`; `environment.txt`); the verifier is self-contained and uses only the Python standard library. **Use of AI assistance:** see Section 8. **Competing interests:** none.

A Certificates

All certificates below are exact rational data; balancing weights are strictly positive multipliers λ with $\sum_S \lambda_S \mathbf{1}_S = \mathbf{1}_N$. This appendix displays the key tables; the *complete* certificate package—per-coalition exactness witnesses for the eight games claimed exact by direct certification (the convex game of Theorem 5.4 is exact by convexity), full satisfaction and excess tables, and balancing weights for *every* cumulative level set of every solution point claimed (including the SD points of the permutation games and the two-player subgame of Theorem 5.4)—is in the machine-readable supplement (`certificates.json`), re-verifiable in exact arithmetic, using only the Python standard library, with the single command `python3 verify_paper.py`.

Theorem 5.1

Satisfaction levels of $\mathrm{SD}(v)$ (ascending; coalitions at each level):

$$\begin{array}{ll} \alpha = 15/4: & \{3\},\ \{2,3\},\ \{1,2,4\},\ \{1,3,4\} \\ \alpha = 53/12: & \{1\},\ \{1,3\},\ \{1,2,3\},\ \{2,4\} \\ \alpha = 19/4: & \{2\},\ \{1,2\} \\ \alpha = 61/12: & \{2,3,4\} \\ \alpha = 205/12: & \{4\},\ \{1,4\},\ \{3,4\} \end{array}$$

The minimal level set is balanced with weights $\lambda_{\{3\}} = \frac{1}{3}$, $\lambda_{\{2,3\}} = \frac{1}{3}$, $\lambda_{\{1,2,4\}} = \frac{2}{3}$, $\lambda_{\{1,3,4\}} = \frac{1}{3}$. For $\mathrm{SD}(v')$ the levels are

$$\begin{array}{ll} \alpha = 29/8: & \{2,3\},\ \{1,2,4\},\ \{1,3,4\} \\ \alpha = 33/8: & \{2\},\ \{3\} \\ \alpha = 35/8: & \{1\},\ \{1,2\},\ \{1,3\},\ \{1,2,3\},\ \{2,4\} \\ \alpha = 43/8: & \{2,3,4\} \\ \alpha = 139/8: & \{4\},\ \{1,4\},\ \{3,4\} \end{array}$$

with minimal level set balanced by $\lambda_{\{2,3\}} = \lambda_{\{1,2,4\}} = \lambda_{\{1,3,4\}} = \frac{1}{2}$. Cumulative lower level sets are balanced at every threshold (weights in the supplement), certifying both points by the criterion of Section 4.

Theorem 5.3

At every sampled point of the segment ($t = -\frac{1}{2}, -\frac{1}{3}, -\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}$, all in the supplement) the bilateral complaints are

$$s_{12} = s_{21} = 2, \qquad s_{34} = s_{43} = 2, \qquad s_{ij} = 0 \text{ for all other ordered pairs,}$$

and the two half-interval stage orders displayed in the proof of Theorem 5.3 extend these equalities to the whole segment.

Theorem 6.1

For the base game K: the nucleolus $(\frac{1}{2}, \frac{33}{4}, \frac{1}{2}, \frac{7}{4})$ has maximal excess 0, attained exactly by the collection $\{\{1,3\},\{2,4\}\}$, balanced with weights $\lambda = 1, 1$ (a partition). For the bumped game K': the nucleolus $(\frac{2}{3}, \frac{23}{3}, \frac{1}{3}, \frac{7}{3})$ likewise has maximal excess 0 attained by the partition $\{\{1,3\},\{2,4\}\}$. Kohlberg's criterion requires every cumulative upper excess set to be balanced; the supplement lists all eight (resp. seven) thresholds with strictly positive weights for both games, together with the complete SD certificates for the SD subclaim.

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